# Thresholds That Travel: Adaptive Alarm Thresholds Let Attacks Through in ICS Anomaly Detection

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## Abstract

Anomaly detectors for industrial control systems (ICS) raise an alarm when a score crosses a threshold. The threshold is calibrated on clean data, and adaptive rules then move it online: daily recalibration, a running quantile, a refitted extreme-value tail. Each rule learns from part of its input, and we ask what happens when attacks land there.

We report three pre-registered studies on six datasets that no earlier analysis of ours had opened; one of their four primary rules passed. Counting an attack as detected when any alarm falls inside it, a plain quantile tracker detected fewer attacks than the frozen calibration threshold on HAI 20.07 (20 of 38 against 36), on BATADAL (6 of 14 against 14) and among the Tennessee Eastman attacks launched directly after another attack (66 of 104 against 104). Exploratory follow-up narrows each result. On Tennessee Eastman every one of those lost attacks lies in a sequence of noise attacks on which the tracker alarmed at once and fell silent a median 24 hours later, while the attack continued. BATADAL's loss disappears when the tracker's step is scaled by the threshold and not by a heavy-tailed spread. On HAI 20.07 the frozen threshold was in alarm for 39% of the unlabelled time, and alarms shifted at random would still have detected 17 attacks. Counting only alarms that begin inside the attack, the clearest loss is on three development releases (76 of 160 against 124, exploratory), and ProvICS shows none under either count.

Poisoning campaigns injected at the score level into 56 sealed attack-free Tennessee Eastman runs (registered) worked against each class of rule. After a day of two-minute pulses the tracker caught none of 56 later bumps just above the calibrated threshold; pulses shorter than a rule's event length did the same to a recalibration that leaves alarm episodes out; and an attacker who read the threshold and held the score just under it left a standard SPOT threshold catching 16 of 56 bumps a quarter above calibration. The frozen threshold caught all 56 each time, and a fixed threshold watching beside the attacked rule would have been in alarm for a median 99% of the silent campaign. A ceiling at $\kappa$ times the calibrated threshold confines these attacks to events at or below it. Automatic tightening cost up to 3.5 times the frozen threshold's false alarms and, on Tennessee Eastman, detected no more than a slightly lower fixed threshold at the same alarm rate. We conclude that a threshold which loosens itself needs a ceiling and a fixed reference threshold beside it, and we release the registrations, ledger and code.

## Keywords

industrial control systems, anomaly detection, adaptive thresholds, poisoning, extreme value theory, pre-registration

## 1 Introduction

A process-monitoring anomaly detector has two parts: a score that grows when the plant behaves unlike its normal operation, and a threshold that turns the score into alarms. Most of the research effort goes into the score. The threshold is usually set once, on held-out normal data, to meet a false-alarm budget, and then expected to hold that budget on a plant that drifts: sensors age, seasons change load, and operators retune loops. When it does not hold, the standard remedies adapt the threshold online. Daily recalibration re-estimates a quantile on the last day of data. Online quantile tracking, the mechanism behind adaptive conformal inference [15], nudges the threshold up after every exceedance and down otherwise. Streaming peaks-over-threshold methods (SPOT and DSPOT [31]) refit a generalised Pareto tail as new values arrive, and the nonparametric dynamic threshold of Telemanom [19] recomputes its level on a trailing window.

Each of these rules learns from part of its input. Trackers and recalibration learn from the values that exceed the threshold. SPOT and DSPOT discard values above their alarm level but learn from values between a base level and the alarm level. A frozen threshold learns from nothing after calibration. We call the part of the score range a rule learns from its *learning region*, and the question of this paper is what happens when an attack, or an attacker, lands in it.

The answer depends on the region. Each class of rule that can raise its own threshold was pushed up through the part of its input it learns from, by recorded attacks or by an injected campaign. How much detection this costs depends on the step size, on how long attacks last and on how a detection is counted, and we report where the effect held and where closer analysis reduced it:

- **A plain tracker stops alarming during a sustained attack and stays raised.** An attack is a long exceedance, so a tracker climbs through it. Counting any alarm inside an attack (registered), the tracker detected 20 of HAI 20.07's 38 attacks against 36 for the frozen threshold, 6 of BATADAL's 14 against 14, and 66 of 104 Tennessee Eastman attacks launched directly after another attack against 104. Follow-up analysis (exploratory) shows what these counts contain. The Tennessee Eastman losses are the later attacks of noise sequences whose start the tracker flagged; BATADAL's loss needs the registered step of 6.9 times the threshold per sample; and on HAI 20.07 the frozen threshold's standing alarms would have collected 17 detections by chance. Two stronger exceedance learners kept every one of those Tennessee Eastman attacks (registered).
- **The same rules can be poisoned on purpose and loudly.** Twelve two-minute pulses a day raised these thresholds enough to hide a later attack below the pulse height that a

frozen threshold caught every time (registered as a score-level injection on sealed attack-free Tennessee Eastman runs; exploratory on HAI).

- **Rules that refuse to learn from alarms can be poisoned without alarming themselves.** An attacker who reads the current SPOT threshold and holds the score just under it feeds every sample into the fitted tail. A standard SPOT then caught 16 of 56 later attacks a quarter above the calibrated threshold, against 56 without the campaign (registered, score level). The campaign is silent only to the rule it attacks: the score sat above the calibrated threshold for a median 99% of it (exploratory).
- **Rules that leave alarm episodes out of their update can be poisoned by alarms too short to count.** Pulses shorter than a rule's event length are not left out, so the rule learns from them (registered, score level; the campaign raises 48 short alarms a day).

A threshold that only tightens itself cannot be pushed above the frozen threshold by any of these, so under the any-alarm count it keeps the frozen threshold's recall by construction (Section 3.1); it paid for that with false alarms and, on Tennessee Eastman, detected no more than a slightly lower frozen threshold. A ceiling at $\kappa\tau_0$ extends the guarantee to rules that loosen: no poisoning can hide an event above the ceiling.

*Method.* Three studies were registered, each before any test row or label of its datasets was read: tt1 (HAI 20.07, HyGrid-ICS, vorausAD), tt1b (BATADAL) and tt1d (Tennessee Eastman, ProvICS; for these the dataset documentation, the file names and the header lines of the label files were read first). A ledger records every access to recorded attack data, the analysis harness refuses to run if any frozen source file changed after registration, and it computes the registered gates itself. The rules and the attack designs came from development on three earlier HAI releases whose attacks had already been used; we report how the claim changed during development, including two hypotheses we dropped. Extensions run after the registered finals are labelled exploratory.

*Contributions.*

(1) Three registered studies of threshold rules on six never-opened datasets (hardware-in-the-loop and simulated plants, an industrial robot), reporting every registered outcome, including three failed primary rules and one dated amendment (Sections 5 and 6).
(2) A registered account of what a plain exceedance-learning threshold does under real attacks: it alarms, climbs and goes quiet while the attack continues. We show how much of its any-alarm losses on HAI 20.07, BATADAL and Tennessee Eastman survives a chance reference, a stricter onset count and a threshold-scaled step (exploratory), and that two stronger rules avoid the loss (registered).
(3) A learning-region analysis of adaptive thresholds: how long the tracker and daily recalibration remember an exceedance (Section 3.1), and a matching score-level attack for each class, registered on sealed attack-free runs: loud pulses for exceedance learners, a closed-loop frog-boiling attack for extreme-value learners, and sub-event pulses for rules that leave alarms out (Section 6).
(4) Design guidance scoped to what was measured: a ceiling bounds any poisoning, a fixed reference threshold exposes the silent campaign, and automatic tightening has a measured cost and no measured benefit (Section 8).
(5) The registration, access ledger, harness and verifier, in the public repository at https://github.com/VyomKulshrestha/Ferrum-OS.

## 2 Background and related work

*ICS anomaly detection.* Process-level detectors learn normal behaviour from sensor and actuator traces and flag deviations: model residuals, autoencoder reconstruction errors, or statistical bounds. Public testbed recordings with labelled attacks (SWaT [16], WADI [1], BATADAL [33], and the HAI releases: HAI 1.0 (20.07) [29], 21.03 [30], later releases in the dataset repository) are the usual benchmarks. Evaluations usually fix one threshold per detector, chosen on clean data, by an offline peaks-over-threshold fit [32], or, less defensibly, on the test labels. Wu and Keogh [36] list further benchmark pitfalls.

*Threshold rules.* Static thresholds meet a quantile or false-alarm budget on clean data. Adaptive variants recalibrate on a trailing window, track a target exceedance rate online [15], fit an extreme-value tail and update it per observation (SPOT, and DSPOT for drifting streams [31]), or recompute a level on a trailing window of smoothed errors [19]. Game-theoretic time-varying thresholds [14] are precomputed, not learned from the stream. Alarm-management practice treats nuisance and chattering alarms as a condition to eliminate, with explicit alarm-rate targets, and routes changes to an alarm through authorised management of change [11, 20]; operators who acknowledge nuisance alarms without investigating them are the failure mode our pulse campaigns exploit.

*Poisoning and stealthy attacks.* Rubinstein et al. [26] showed that slowly growing ("boiling frog") poisoning of a weekly retrained PCA detector, whose threshold is re-estimated with it, can stop it flagging a later attack. Kloft and Laskov [21, 22] showed that bounding the false-positive rate limits poisoning of online centroid detectors. Chan-Tin et al. [6, 7] described frog-boiling against outlier rejection in network coordinate systems: every update stays within the rejection threshold while the system drifts. In ICS, Kravchik and Shabtai [24] found an autoencoder on SWaT physical-state data resistant to poisoning, but later work poisoned online-retrained ICS detectors with sub-threshold samples and proposed lowering the threshold as mitigation [23], and Seo et al. [27] poisoned a drift-adaptive SWaT detector through its retraining step, evading its drift detector until the retrained threshold cut precision from 0.91 to 0.24. Erba et al. [12] concealed attacks from reconstruction-based detectors by manipulating sensor readings. Attackers who hold a detection statistic just under a fixed threshold are standard in control security [5, 34]. Hong [17] studied gradual drift against world-model monitors in simulation, without attacks or adaptive thresholds. These works target the model or the score, or assume a fixed threshold. We target the

adaptive threshold rule and show that it can fail even when the model is not poisoned.

*Distribution shift and calibration.* Adaptive conformal inference and its successors keep a long-run miscoverage rate under arbitrary, even adversarial, sequences [3, 15], a guarantee about the rate of exceedances. Atarmla [2] proves that for the online quantile tracker this guarantee is an accounting identity and bounds the evidence a step or ramp produces before the tracker absorbs it. The guarantee holds whether the exceedances come from benign drift or from attacks, the property we exploit in Section 7; Proposition 1 (Section 3.1) follows from the same identity, which we test on recorded attacks, and we extend the question to extreme-value thresholds.

## 3 Threat model and learning regions

*System.* A monitor computes a score $s_t$ from plant telemetry and raises an alarm when $s_t > \tau_t$. The rule that sets $\tau_t$ is calibrated once on clean data to a false-alarm budget and then runs online. Operators acknowledge alarms; persistent nuisance alarms are acknowledged without investigation.

*Learning region.* For a rule $R$, the learning region $L(R, \tau_t)$ is the set of score values that change $R$'s future thresholds. Table 1 gives the rules we study.

A rule can loosen (raise $\tau_t$) only through values in the part of its learning region that pushes upward. That gives three attacker classes, from weakest to strongest:

- **A0, no attacker.** Real attacks are themselves exceedances. A rule whose upward learning region includes exceedances is poisoned by the attacks it is meant to detect.
- **A1, open-loop injection.** The attacker can perturb one control loop (setpoints or sensor readings) without seeing the monitor. Supra-threshold pulses land in the exceedance region; sub-alarm pulses land in the band.
- **A2, closed-loop.** The stealthy attacker of control security [5, 34], turned against an adaptive threshold: it can read $\tau_t$ (white-box, or by probing) and shape the score. Holding $s_t$ at $0.97\,\tau_t$ places every sample at the top of the band with no alarm. We evaluate A2 at the score level, an upper bound on what a physical attacker can do, and through noise spoofed onto three sensor readings of one loop, scored by the real monitor.

The attacker's goal is a terminal attack that the frozen threshold would detect but the adapted threshold does not. Table 2 lists what each registered campaign needs.

*Why poison rather than conceal.* An A2 attacker who can spoof sensors could also try to hide the terminal attack directly, by spoofing the readings it disturbs. Poisoning moves that work earlier: after the campaign the terminal step runs unmodified, so the attacker does not have to keep every reading it changes consistent with normal operation, which constrained-concealment attacks must do [12]. We did not compare the two strategies beyond the replay baseline of Section 7.5, which hid more than any campaign. The A1 assumption that operators acknowledge recurring alarms without investigating is the condition alarm-management standards try to remove [20]; our pulse campaign adds twelve pulses a day, and the frozen threshold alarmed on 92% of them on HAI 20.07 (registered E4).

### 3.1 How long a threshold remembers

Three facts follow from the update rules alone. They predict most of what Sections 5 and 7 measure. Proposition 1 restates, as a memory, the counting identity behind the long-run coverage guarantee of adaptive conformal inference [15], from which Atarmla [2] derives the excess alarms a persistent step produces before the tracker absorbs it. What we add is its reading in terms of the calibrated false-alarm rate and its check on recorded attacks.

*Proposition 1 (tracker memory). Under $q_{t+1} = q_t + \eta(\mathbf{1}[s_t > q_t] - p)$, each step with $s_t > q_t$ raises the threshold by $\eta(1-p)$ and each other step lowers it by $\eta p$. A rise built from $E$ exceedance steps therefore takes $E(1-p)/p$ quiet steps to undo, whatever the step size $\eta$.*

The calibrated exceedance rate $p$ sets the memory. On HAI 20.07 ($p = 7.2 \times 10^{-4}$ per second) the ratio is 1,385, so one minute above the threshold keeps it raised for 23 hours. On BATADAL ($p = 0.0075$ per 15-minute step) it is 133, so one step above the threshold keeps it raised for 33 hours. The step size sets how high an exceedance lifts the tracker, not how long the lift lasts: on BATADAL, where $\eta$ is $6.9\,\tau_0$ per step, one exceedance step lifts it by almost seven times $\tau_0$. A slower tracker (tracker_slow, $\eta/10$) climbs less during a short attack but needs the same 1,385 quiet seconds per exceedance second to come back down, which is why it lost recall too (Section 5.2). Recovery time bounds blind time from above: the tracker misses a later attack only while it sits above that attack's score. Post-hoc checks on the recorded trajectories agree with the proposition: every change of the tracker on the HAI 20.07 and BATADAL test streams is a whole number of these up and down steps (largest deviation $7 \times 10^{-6}$ of a step), and from its final levels the tracker would have needed 43 to 209 days of exceedance-free data to return to $\tau_0$.

*Proposition 2 (daily memory). Daily recalibration sets the next day's threshold to the $(1-p)$ quantile of the last day's $N$ steps. An attack that holds the score above a level $a$ for more than $pN+1$ steps of one day lifts the next day's threshold to at least $a$.*

That is 64 seconds on HAI 20.07 and two 15-minute steps on BATADAL. The memory is one day, so daily recalibration is poisoned whenever attacks fall on consecutive days, as they do in the HAI test files, and it still catches an attack whose first day follows a quiet one. On BATADAL, whose attacks last one to five days and begin 1.7 to 24 days after the previous one ends (post-hoc, from the recorded labels), it caught 14 of 14 with its threshold at $0.69\,\tau_0$ at the median onset (registered tt1b).

*Proposition 3 (tighten-only). If $\tau_t \le \tau_0$ at every step (tighten-only without a loosening permit), every sample that alarms under the frozen threshold also alarms under $\tau_t$. The rule therefore detects every event the frozen threshold detects when any alarm inside the event counts, on any input and against any attacker, adaptive ones included. What it can lose is false alarms. Under the stricter onset count of Section 5.2 the guarantee need not hold, because a lower threshold can start an alarm episode before an attack begins.*

**Table 1: Update and learning region of each threshold rule.**

| Rule | Update | Learning region |
|---|---|---|
| frozen, static margin | none | empty |
| symmetric tracker $q_{t+1} = q_t + \eta(\mathbf{1}[s_t > q_t] - p)$ | up on exceedance, down otherwise | all values; upward moves only from $s_t > \tau_t$ |
| daily recalibration | $(1-p)$ quantile of the last day | the top $p$ of each day, whatever its source |
| SPOT | adds $s_t - t$ to the tail fit when $t < s_t \le z_q$ | the band $(t, \tau_t]$ |
| DSPOT | as SPOT on $s_t$ minus a moving mean of non-alarm values | the band, plus every non-alarm value through the mean |
| tighten-only | min(ceiling, tracker); the ceiling rises only through a permit | downward only, unless a permit fires |

**Table 2: What each registered campaign needs and what it shows the operator (Section 6.3). Every campaign writes the score directly, an upper bound on an attacker who must act through sensors or setpoints.**

| Campaign | Reads | Writes, for 24 h | Alarms raised |
|---|---|---|---|
| loud pulses | $\tau_0$, once | 2 $\tau_0$ for 119 s every 2 h | 12 a day |
| sub-event pulses | $\tau_t$ at each pulse | 2 $\tau_t$ for 43 s every 30 min | 48 a day, each shorter than an event |
| frog | $\tau_t$ at every step | 0.97 $\tau_t$ throughout | none in 51 of 56 placements |

*Proposition 4 (ceiling). If $\tau_t \le \kappa\tau_0$ at every step, every sample above $\kappa\tau_0$ alarms, so under the any-alarm count the rule detects every event that the static threshold $\kappa\tau_0$ detects, on any input and against any attacker.* Proposition 3 is the case $\kappa = 1$. A ceiling bounds poisoning: whatever the stream held before, an attacker can hide only events whose score never exceeds $\kappa\tau_0$.

Proposition 3 means the tighten-only results below measure its cost, not its recall. On HAI 20.07, HyGrid-ICS and voraus-AD its recall and false alarms equalled the frozen threshold's: the inflated tracker under the cap kept it at $\tau_0$ (post-hoc: on HAI 20.07 it sat below $\tau_0$ in 0.4% of minutes). Its own behaviour shows on BATADAL, where the tracker sat below $\tau_0$ for 3.2% of the test time (post-hoc) and the rule ran at 2.43 times the false-alarm budget against 0.70 for frozen (registered), and on a development split of HAI 23.05 (5.98 against 4.05, exploratory). A loosening permit breaks the guarantee, even one that only an operator could grant: with the registered oracle permit, a statistical permit that a simulated operator blocks whenever the review window holds a labelled event, the rule caught the terminal step in 1 of 10 control placements on HAI 20.07, against 10 of 10 for frozen.

## 4 Monitor, rules and data

*Monitor.* D1 is the world-model residual monitor of earlier registered studies on HAI [25]: an action-conditioned ridge regression with LightGBM residual corrections (60 trees, history 3) predicts each sensor one and five steps ahead; the score is the mean of the three largest standardised absolute residuals at each horizon, summed and smoothed over 60 s. D0 is a model-free control: the mean of the three largest standardised first differences. Both are calibrated on the most recent 20% of normal training data: to 2 false-alarm episodes per day on the 1 Hz stream datasets (episodes merged across 60 s gaps), to a per-sample false-alarm target of 0.05 on voraus-AD, and to one false-alarm episode per week on BATADAL (episodes merged across 1 h). In tt1d one monitor is fitted per operating mode of Tennessee Eastman, calibrated on four of that mode's eleven development runs, and one for ProvICS; every duration is converted to the dataset's step; and a third score is a registered secondary: D2, the mean of the three largest standardised residuals after projection on the principal components that explain 95% of the variance.

*Rules.* frozen ($\tau_0$ at the budget); margin ($\tau_0$ + 0.25× the spread between the calibration 99.9th percentile and median); daily (trailing-day quantile at the calibration exceedance rate $p$); tracker ($\eta$ = 0.01× spread on the stream datasets, 0.05× spread on voraus-AD) and tracker_slow ($\eta/10$, the best false-alarm adherence among trackers in development); tighten-only with four automatic permits and an oracle permit that simulates an operator who refuses to loosen while an event is under way: eleven rules in all. On voraus-AD the spread is the calibration 99th percentile minus the median and daily recalibration happens every 50 samples. The streaming extreme-value rules (SPOT and DSPOT, with method-of-moments and maximum-likelihood tail fits [18, 31]) were not part of the tt1 registration: they enter in Section 7 as exploratory and post-hoc analyses and are registered in the BATADAL replication (tt1b) and, in their standard maximum-likelihood form, in tt1d.

*Registration.* The protocol fixed the monitor, splits, rules, injection designs, gates and seeds; the harness recorded one ledger entry per dataset before reading a test array, refused to run if a frozen file's SHA-256 differed from the registration, and computed the gates from its own output. A verifier checks that each access followed the registration commit, that the ledger holds no access the protocol does not allow, and that the stored gates match a recomputation (27 of 27 checks pass for tt1, 8 of 8 for tt1b, and 19 of 19 and 22 of 22 for the two datasets of tt1d, whose verifier also checks that later, exploratory accesses are marked as such in the ledger).

**Table 3: Datasets. Event counts are the labelled events of each final split.**

| Dataset | System | Training data | Final data | Events |
|---|---|---|---|---|
| HAI 20.07 (primary) | HIL testbed (boiler, turbine, water treatment) | 547,624 s normal, 31 sensors, 20 actions | test1 + test2, 5.15 days | 38 attacks |
| HyGrid-ICS | simulated hybrid grid SCADA, 5 Hz to 1 Hz | 9.6 h (label-blind first 40%) | 14.5 h | 91 attacks |
| voraus-AD | industrial robot, 100 Hz to 10 Hz | 948 normal samples | 1174 samples | 755 anomalous samples |
| HAI 21.03, 22.04, 23.05 (development) | HIL testbed | normal training data | spent test files | 50, 58, 52 attacks |
| BATADAL (registered replication) | water distribution network, 15 min | 384 days normal | two test files, 297 days | 14 attacks |
| Tennessee Eastman (tt1d) | simulated chemical process, 7 modes, 3.6 s | 11 attack-free runs of 120 h per mode | 142 attack runs; 56 sealed attack-free runs | 245 attacks |
| ProvICS (tt1d) | reactor twin under a real PLC, 2 s | 39 h attack-free | 21 h attack recording; 13 h sealed attack-free | 32 events |

Each final was launched in the session that committed its registration: the first access follows the commit by 43 s (tt1), 86 s (tt1b) and 28 s (tt1d), and the amended attempt follows its amendment by 17 s. The history shows the order of the two and nothing about how long a plan stood; what stops the analysis from changing after the data were seen is the frozen-file check. tt1d keeps two ledger entries per dataset, for its attack data and its sealed attack-free data; ProvICS has three of each: a failed first attempt, a count of missing values, and the amended attempt (Section 6.4).

*Development.* The registered claim of tt1 differs from the one we set out to test. We began with a tighten-only threshold that loosens through a statistical permit, moved to a static safety margin, and registered frozen against adaptive after the first two lost recall or missed the budget on the development releases (Appendix D). After the third study, two mock reviews and a code audit led to a last exploratory round under its own written plan (tt1f, Appendix F); its results are marked exploratory wherever they appear.

## 5 Registered results

### 5.1 Primary: HAI 20.07

False-alarm episodes per day are counted over the 4.94 days of unlabelled time inside the 5.15-day test streams. G1 passes as registered, but on this stream the pulses were not needed: the adaptive rules also missed the terminal step in the control runs, because the test stream's real attacks had already raised their thresholds 7 to 26 times above $\tau_0$. The pulse effect in isolation comes from development on attack-free normal data (frozen 30/30; daily 4/30, control 21/30; tracker 1/30, control 23/30; tracker_slow 3/30, control 25/30). Neither registered test of a deliberate campaign, G1 here or H3 on BATADAL (Section 6.1), could show an effect, because the rules under test were already blind in the controls; the campaign results in Section 7 are exploratory, and those of Section 6.3 registered.

### 5.2 Real attacks inflate adaptive thresholds

At HAI 20.07 attack onsets the median threshold stood at 25.6 $\tau_0$ for daily recalibration, 12.9 for the tracker and 6.4 for the slow tracker (Figure 1). HyGrid's daily rule equals frozen because its 14.5 h final never completes a day.

On HAI 20.07 and BATADAL every disagreement between an adaptive rule and the frozen threshold went one way. On HAI 20.07 the frozen threshold caught 16 attacks the tracker missed and the tracker caught none that it missed (exact McNemar $p = 3.1\times10^{-5}$); the slow tracker split 9 to 0 ($p = 0.0039$) and daily recalibration 18 to 0 ($p = 7.6 \times 10^{-6}$). On BATADAL the tracker split 8 to 0 ($p = 0.0078$). These counts are post-hoc, from the recorded trajectories, which reproduce every registered event count.

*Time in alarm and onset recall (post-hoc).* Episode counts hide how long an alarm lasts. On HAI 20.07 the frozen threshold was in alarm for 39% of the unlabelled time. The D1 score sat above $\tau_0$ for the first 23.8 h of test1, of which the first 2.7 h carry no attack label and seven attacks begin inside the rest, and for a 17-hour stretch of test2 that opens with an attack and holds two more. An alarm episode that overlaps an attack does not count as a false alarm, so the registered episode count records none of these hours. Ten of the frozen threshold's 36 detections were attacks that began while an alarm was already running, nine of them inside these two runs. Counting an attack only when an alarm episode starts inside it, the detection rule that the authors of the Sherlock dataset recommend [35], the frozen threshold detects 26 of 38, the tracker 20 ($-0.16$, $[-0.37, 0.05]$), the slow tracker 27 ($+0.03$, $[-0.18, 0.24]$) and daily recalibration 11 ($-0.39$, $[-0.55, -0.24]$). On BATADAL the frozen threshold was in alarm for 1.4% of the unlabelled time and the stricter count changes nothing (tracker $-0.57$, $[-0.79, -0.29]$). On HyGrid-ICS (4.1% in alarm) the 91 attacks cover 83% of the final rows, about 100 s apart on average, so one merged alarm episode spans several attacks and the stricter count credits few of them: frozen 3 of 91, tracker 8 ($+0.05$, $[0.00, 0.11]$), slow tracker 5. There the stricter count reverses the sign, but it measures how alarms merge across back-to-back attacks more than what the threshold detects. G2 stands as registered, but the tracker's HAI 20.07 gap depends on crediting the frozen threshold's long alarms. A chance reference says how much (exploratory, from the per-minute trace of the final): shifting each rule's alarm sequence against the labels

**Table 4: Registered gates of tt1 on HAI 20.07. Data status: registered.**

| Gate | Registered test | Result |
|---|---|---|
| G1 | after a pulse campaign ($12 \times 120$ s per day at twice the terminal magnitude), daily, tracker and tracker_slow detect the terminal step in at most (frozen $-$ 0.5) of placements; frozen at least 0.8 | frozen 10/10, each adaptive rule 0/10: pass, but uninformative (the adaptive controls were already 0/10) |
| G2 | recall(tracker) $-$ recall(frozen) on the real attacks, 95% paired event bootstrap upper bound below 0 | 0.526 vs 0.947, $-0.42$ $[-0.58, -0.26]$: pass |
| G3 | frozen false-alarm episodes per day on unlabelled test time at most 4 | 2.23: pass (above the 2.0 budget; in alarm for 39% of the unlabelled time, post-hoc) |

**Table 5: Event recall and the paired difference against the frozen threshold (95% paired event bootstrap) on the D1 score. Data status: registered (tt1).**

| Dataset (events) | frozen | daily | tracker | tracker_slow | tighten-only |
|---|---|---|---|---|---|
| HAI 20.07 (38 attacks) | 0.947 | 0.474 $[-0.63, -0.32]$ | 0.526 $[-0.58, -0.26]$ | 0.711 $[-0.37, -0.11]$ | 0.947 |
| HyGrid-ICS (91 attacks) | 0.297 | 0.297 | 0.154 $[-0.22, -0.08]$ | 0.165 $[-0.21, -0.07]$ | 0.297 |
| voraus-AD (755 anomalous samples) | 0.334 | 0.148 $[-0.21, -0.16]$ | 0.082 $[-0.28, -0.22]$ | 0.213 $[-0.14, -0.10]$ | 0.334 |

**Table 6: False alarms. Data status: registered (tt1).**

| Dataset | frozen | margin | daily | tracker | tracker_slow |
|---|---|---|---|---|---|
| HAI 20.07 (episodes/day) | 2.23 | 2.23 | 0.20 | 1.01 | 2.23 |
| HyGrid-ICS (episodes/day, 2.5 h unlabelled) | 19.2 | 0.0 | 19.2 | 19.2 | 19.2 |
| voraus-AD (per-sample FPR, target 0.05) | 0.081 | 0.033 | 0.086 | 0.012 | 0.014 |

by a random offset detects 17 of the 38 attacks on average for the frozen threshold (central 95% range 11 to 23) and 1.4 for the tracker, so net of chance the two rules are close. The BATADAL gap does not depend on long alarms; it depends on the tracker's step (Section 6.1).

## 5.3 False alarms

Where an adaptive rule raised fewer false alarms than frozen, it bought them with the recall losses of Section 5.2, except for the oracle permit on voraus-AD (0.064 against 0.081 at equal recall), which uses the attack labels. The one rule that cut false alarms at no recall cost here, the static margin on HyGrid-ICS (0.0 against 19.2 per day at identical recall), does not adapt. The margin matched frozen recall on HAI 20.07 and held voraus-AD's target at a cost of 12 points of recall there. Counted in time rather than episodes (post-hoc), the frozen threshold was in alarm for 39% of the unlabelled time on HAI 20.07, 4.1% on HyGrid-ICS and 1.4% on BATADAL; the tracker for 0.2%, 0.3% and 0.4%. Tighten-only equalled the frozen threshold on all three tt1 datasets (2.23, 19.2 and 0.081).

## 5.4 Other registered outcomes

The protocol commits us to report every rule, gate and failure; Appendix A gives the rest of the registered record. Every automatic loosening permit lost recall on HAI 20.07, and the oracle permit, which equalled frozen on the real attacks, caught the terminal step in 1 of 10 control placements of the G1 campaign.

# 6 Registered replications

## 6.1 BATADAL (tt1b)

We registered a second study (tt1b) to test the exploratory findings of Sections 7.1 and C.2 on data no analysis here had opened: the C-Town network of BATADAL [33], whose original attacks conceal themselves in the SCADA readings, in the re-simulation without that concealment published with the artifact of Erba et al. [12], put on a 15-minute grid. All time-scale settings were fixed in physical time before access: 1 h smoothing, episodes merged across 1 h, one false-alarm episode per week, a one-day DSPOT mean. The test files hold 14 attacks in 297 days, 260 of them unlabelled.

We retain the primary rule as registered and report H3 as uninformative rather than refuted. H2 passes under the registered bootstrap, whose interval is $[0, 0]$ because no event was discordant; an exact one-sided 97.5% bound for 0 discordant events in 14 is $-0.23$, beyond the $-0.14$ tolerance, so H2 is not informative under an exact method. SWaT and WADI are registered under the same protocol. We requested them from their publisher on 2026-10-23 and had not been granted access when we submitted, so the registered replication rests on BATADAL alone; if access arrives, they will run once under the registration and be reported whatever they show.

H3 could not be tested because DSPOT's control was already blind, and the blindness predates the attacks. Fitted by the method of moments to BATADAL's heavy-tailed calibration scores (spread 6,462 against $\tau_0 = 9.39$), SPOT's alarm level started at 49.0 $\tau_0$ and DSPOT's at 45.8 $\tau_0$ (post-hoc trajectories), and MLE SPOT at 1.22 $\tau_0$. The registered frog variants are in Appendix A.

**Table 7: Registered hypotheses of tt1b on BATADAL. Data status: registered.**

| Hypothesis | Registered test | Result |
|---|---|---|
| H1 | recall(tracker) − recall(frozen), interval upper bound below 0 | 6/14 vs 14/14, −0.57 [−0.79, −0.29]: pass |
| H2 | recall(MLE SPOT) − recall(frozen), lower bound at least $-\max(0.10, 2/N)$ | 14/14, difference 0: pass |
| H3 | 24 h closed-loop frog: DSPOT at least 0.5 below its control; frozen and tighten-only within 0.1 | DSPOT already 0/10 in the control: fail |
| Primary rule | H1 and H2 and H3 | fail |

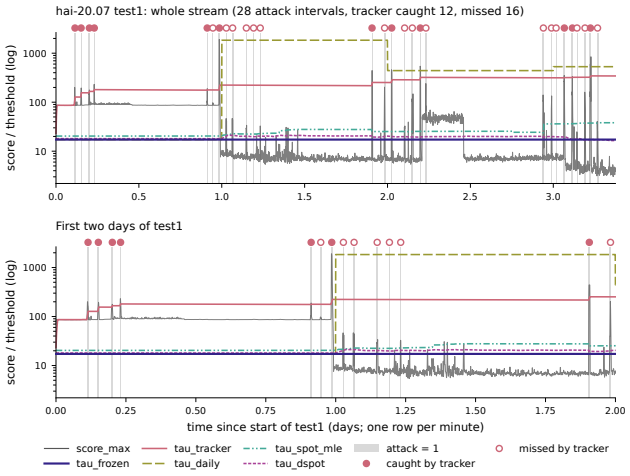



**Figure 1: Threshold trajectories on the HAI 20.07 test1 stream (4,860 minutes). Grey: the D1 score score_max. Coloured: the frozen, tracker, daily, spot_mle and dspot thresholds, log scale. Grey shading marks attack-labelled minutes; each marker above the panel is one attack interval, filled if the tracker caught it. test1 holds 28 of the 38 attacks; the tracker caught 12. The score stays above $\tau_0 = 17.26$ for the first 23.8 h of test1, so the frozen threshold is in alarm throughout; only the first 2.7 h carry no attack label, and seven attacks begin inside the run. That unlabelled opening lifted the tracker to $5.0\,\tau_0$ before the first attack ($5.25\,\tau_0$ at its onset), and the early attacks then raised the tracker and daily thresholds far above frozen. Later attacks arrive under the inflated thresholds. Data status: post-hoc analysis; the trajectories were computed after the registered finals. The recomputed calibration threshold equals the registered one exactly. Over test1 and test2 the traces give frozen 36/38 and tracker 20/38 (registered tt1) and spot_mle 35/38 and dspot 36/38 (post-hoc: SPOT and DSPOT were not in the tt1 registration). test2 is not drawn; test1 has no missing values.**

*The tracker's step (exploratory).* The same heavy tail set the registered tracker's step, 0.01 of the spread, to $6.9\,\tau_0$ per 15-minute sample. Rerun from the stored trace of the registered final with the step of tt1d's second tracker, $0.01\,\tau_0$, the tracker detected all 14 attacks under both counts at 0.30 times the false-alarm budget, and also under ceilings of 1.1 and $1.25\,\tau_0$; the excluding recalibration and the capped tracker detected all 14 as well (Table 8). H1 stands as registered, but the BATADAL loss belongs to a step scaled by a heavy-tailed spread and is no evidence against quantile tracking as such. The registered tracker's 6 detections are what its alarms give by chance (5.8 on average under random shifts), and daily recalibration's 14 of 14, at 6.34 times the budget, sit inside its chance range of 10 to 14.

**Table 8: BATADAL from the stored trace of the registered final: the stronger rules and a tracker whose step is $0.01\,\tau_0$. Durations scaled as in tt1b (an event is 1 h). Data status: exploratory (tt1f); the registered counts reproduce (frozen 14, tracker 6).**

| Rule | any / onset (of 14) | FA (× budget) | Onset infl. |
|---|---|---|---|
| tracker, $\eta = 0.01\,\tau_0$ | 14 / 14 | 0.30 | 3.13 |
| daily, alarm episodes left out | 14 / 14 | 0.70 | 1.00 |
| tracker, rise stopped after an event | 14 / 10 | 3.05 | 1.27 |
| tracker, rise capped at $0.1\,\tau_0$ a day | 14 / 14 | 0.35 | 1.61 |
| static $1.1\,\tau_0$ | 14 / 14 | 0.51 | 1.10 |
| static $1.25\,\tau_0$ | 14 / 14 | 0.43 | 1.25 |
| registered tracker, ceiling $1.1\,\tau_0$ | 14 / 13 | 42.56 | 0.79 |
| registered tracker, ceiling $1.25\,\tau_0$ | 14 / 13 | 35.87 | 0.92 |
| tracker, $\eta = 0.01\,\tau_0$, ceiling $1.1\,\tau_0$ | 14 / 14 | 0.59 | 1.02 |
| tracker, $\eta = 0.01\,\tau_0$, ceiling $1.25\,\tau_0$ | 14 / 14 | 0.43 | 1.17 |

With the model-free D0 score, MLE SPOT also lost recall (9/14, [−0.64, −0.14]). The margin and the tracker step are scaled by the calibration spread (99.9th percentile minus median); BATADAL's residual scores are heavy-tailed, so the margin sat 173 times above $\tau_0$. Parameters scaled by a tail spread are fragile when the tail is heavy.

## 6.2 Tennessee Eastman: first attacks and followers (tt1d)

A third study registered what the first two left open: both detection counts, the mechanism, the stronger rules and ceilings of Sections C.3 and C.4, and campaigns whose controls can pass or fail. Its main dataset is Kaspersky's Tennessee Eastman attack set [10,

**Table 9: BATADAL, D1 score. Data status: registered (tt1b).**

| Rule (D1) | Recall | False alarms (× budget) | Onset inflation |
|---|---|---|---|
| frozen | 14/14 | 0.70 | 1.0 |
| tighten-only | 14/14 | 2.43 | 1.0 |
| SPOT (MLE) | 14/14 | 0.81 | 1.0 |
| daily | 14/14 | 6.34 | 0.69 |
| tracker | 6/14 | 1.83 | 409 |
| tracker_slow | 5/14 | 0.70 | 74 |
| SPOT (moments) | 6/14 | 0.13 | 26 |
| DSPOT | 6/14 | 0.13 | 24 |
| margin | 3/14 | 0.05 | 173 |

**Table 10: Tennessee Eastman, D1: events detected, and false-alarm episodes over the budget on the 56 sealed attack-free runs (280 days). Data status: registered (tt1d).**

| Rule | First (141) | Followers (104) any alarm | Followers (104) onset | False alarms |
|---|---|---|---|---|
| frozen | 116 | 104 | 44 | 0.98 |
| tracker | 116 | 66 | 37 | 1.21 |
| tracker, $\eta = 0.01\,\tau_0$ | 123 | 55 | 37 | 1.64 |
| tracker_slow | 110 | 88 | 50 | 0.83 |
| daily | 106 | 59 | 40 | 1.22 |
| daily, episodes left out | 113 | 104 | 44 | 1.33 |
| tracker, capped rise | 130 | 104 | 44 | 2.12 |
| tracker, ceiling $1.25\,\tau_0$ | 121 | 104 | 41 | 1.30 |
| SPOT (ML) | 106 | 104 | 43 | 0.84 |
| tighten-only | 122 | 104 | 44 | 1.57 |

13]: 142 attack runs in seven operating modes (one holds no labelled attack) and attack-free runs of 120 h, one row every 3.6 s. File names fixed each run's role before any content was read. Per mode, eleven attack-free runs fit and calibrate one monitor and eight more stayed sealed. Two attack types hold one attack per run. The other two hold several in a row on different variables, three denial-of-service attacks or four noise attacks in the dataset's description and two or three events here, because back-to-back attacks with the same indicators merge under our event definition. An event is either the *first* of its run (141 events) or a *follower* (104: 45 denial-of-service, 59 noise). Every follower begins within one step of the end of the attack before it, so a follower is the next attack of an uninterrupted sequence.

*H1a passed.* The tracker caught 66 of the 104 followers against 104 for the frozen threshold ($-0.37$, $[-0.47, -0.27]$, whole runs resampled; all 38 discordant events favour frozen), and daily recalibration 59 ($[-0.55, -0.32]$). *H1c passed*: on first events the tracker caught 116 against 116 ($[-0.05, +0.05]$; six discordant events each way). The tracker's loss falls on followers and not on the first attack of a run, the mechanism of Section 7, here registered. Split by attack type (descriptive), all of it sits in the noise sequences: the tracker caught 21 of their 59 followers against 59, and all 45 denial-of-service followers, as the frozen threshold did. In every noise sequence the tracker alarmed on the first event, which lasts

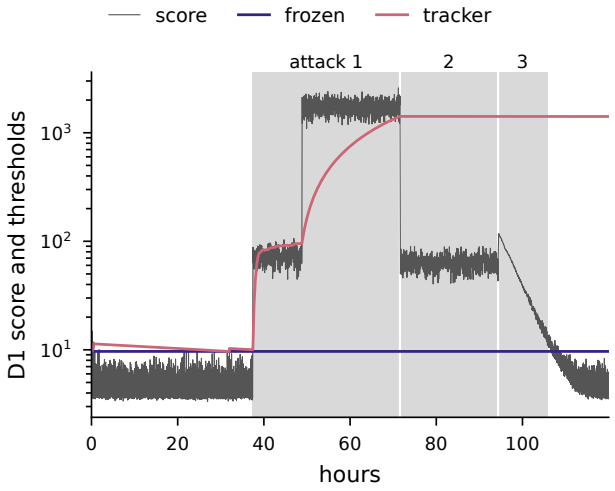



**Figure 2: One Tennessee Eastman run with three consecutive noise events (shaded). The frozen threshold is in alarm for at least 99.99% of each event. The tracker alarms at the start, climbs to $146\,\tau_0$ during the first event (in alarm for 70% of it), and is in alarm for 0.04% of the second and none of the third. The run was fixed in the plan of the exploratory round before any score was looked at. Data status: exploratory, a second, ledgered access after the registered final.**

a median 25 h, and it stood at a median $82\,\tau_0$ (at least 23) when the followers began. What it lost is the alarm on the later attacks of a sequence whose start it had flagged. In these sequences (exploratory; median length 41 h) the frozen threshold and the tracker alarmed in every sequence, at a median of one step after its start; the frozen threshold then stayed in alarm, and the tracker's last alarm came a median 23.7 h in (Figure 2). At the level of runs the two rules are equal: each detected at least one attack in 128 of the 141 runs. Daily recalibration also lost first events (106 against 116, $[-0.12, -0.02]$). *H1b failed*: under the onset count the tracker caught 37 followers against 44 ($-0.07$, $[-0.13, 0.00]$, the upper bound exactly zero), so the registered primary rule, H1a and H1b, failed. The registration had named the reason in advance: a follower begins while the alarm of the attack before it is still running, and the onset count then credits the frozen threshold with only 44 of the 104 followers it alarmed on (6 of the 59 noise followers, against 1 for the tracker). *H2 passed*: the excluding recalibration and the capped tracker each caught all 104 followers. H3, the ceiling's consistency with Proposition 4, held on every event. With D0 (secondary) the tracker again lost followers and not first events ($[-0.30, -0.10]$ and $[-0.06, +0.07]$). With D2 it tightened instead: on the unlabelled time of the attack runs it ran at 33 times the false-alarm budget, against 1.7 for frozen, and caught 134 first events against 79, with a follower difference of $[-0.11, +0.03]$. Appendix E lists every registered rule.

*Chance reference (exploratory).* Tennessee Eastman attacks last from a minute to more than a day, and a rule that alarms a few times a day will often alarm inside a long one. Shifting each rule's

alarm sequence by a random offset within its run gives the frozen threshold 61 of the 141 first events on average (central 95% range 51 to 69) against 116 observed, and the tracker 56 against 116. The integrity attacks are detected far above chance (34 of 34 against 0.3). The 10-hour denial-of-service attacks are not: the frozen threshold's 28 of 41 compare with a chance range of 12 to 24, and the capped tracker's 34 with 28 to 37. About half of the first-event detections in Table 10 are what the alarm rates alone would give, which weakens H1c and the higher first-event counts of the rules that run above the budget. A frozen threshold lowered by 1.5 to 6% to match each of those rules' alarm rates on the sealed runs detected as many first events or more (Tables 11 and 12).

## 6.3 Tennessee Eastman: registered campaigns

Each of the 56 sealed attack-free runs received one campaign placement, drawn after its first 12 h: a 24 h campaign at the score level, one quiet hour, then a 600 s terminal bump (Table 13). The loud campaign sets the score to $2\,\tau_0$ for 119 s (33 steps) every 2 h; the frog and the sub-event pulses are those of Section C.4, with every duration rounded to the 3.6 s step (a sub-event pulse lasts 43 s).

All three campaign hypotheses passed. *H4*: after the frog the standard SPOT caught 16 of 56 bumps of $1.25\,\tau_0$ against 56 in its control ($[-0.82, -0.59]$), and with a ceiling at $1.25\,\tau_0$ all 56 bumps of $1.5\,\tau_0$. In the rehearsal SPOT's own level started up to $1.06\,\tau_0$, which is why H4 was registered at $1.25\,\tau_0$ and not $1.05\,\tau_0$. The 56 placements are seven monitor-level outcomes: the eight placements of a mode always agree, and SPOT caught the $1.25\,\tau_0$ bump in two modes and missed it in five. In three modes its threshold ran away numerically (medians of $10^8$ to $10^{18}\,\tau_0$), and the fitted shape left its attack-free range of $-0.16$ to 0.35 in 55 of the 56 placements. Bounding the shape to 0.5 removed the runaway and blinded the rule more evenly (0 of 56 up to $1.5\,\tau_0$); runs declustering resisted this continuous campaign (56 of 56 from $1.25\,\tau_0$; both exploratory, Appendix F). The campaign is silent only to the rule it attacks: a fixed $\tau_0$ monitor reading the same score would have alarmed in all 56 placements, within 2 h, and stayed in alarm for a median 99% of the campaign (exploratory). *H5*: after the loud pulses the tracker caught none of 56 bumps of $1.05\,\tau_0$ against 45 in its control, and the excluding recalibration was unchanged (49 against 49). At $2\,\tau_0$, the height of the pulses, the tracker caught all 56 again, so this campaign hides only attacks below its own pulse height from the tracker; daily recalibration missed all 56 there too. *H6*: the sub-event pulses left the excluding recalibration and the hold-off tracker with none of 56, against 49 and 47, while both ceilings caught all 56 bumps of $1.5\,\tau_0$ (in 4 of the 56 placements an alarm episode of the excluding rule still reached event length). The frozen threshold caught every bump in every campaign and control. The quiet-log approval gate (secondary) caught 1 of 56 bumps of $1.05\,\tau_0$ after the frog with a review every 6 h and 32 with a daily review, against 44 in its control. Over the 280 sealed attack-free days the frozen threshold ran at 0.98 times its false-alarm budget (Table 10); most adaptive rules ran above it, from 1.21 for the tracker to 2.12 for the capped tracker, and SPOT (0.84) and the slow tracker (0.83) below.

## 6.4 ProvICS (tt1d)

The second dataset of tt1d is ProvICS [28]: a chemical reactor twin under a real PLC, with 32 labelled events in a 21 h recording. The last quarter of its attack-free recording was sealed. The registered run stopped with an error before any result was printed or written, because one sealed row lacks its setpoint values; a ledgered read then counted missing values and nothing else, and a dated amendment, committed before the second attempt, repeats the previous value in an empty cell and changes nothing else.

*P1 failed*: the tracker detected exactly the events the frozen threshold detected, 25 of 32 under the any-alarm count and 4 under the onset count. The monitor did not hold its calibration on this recording. The frozen threshold was in alarm for 44% of the unlabelled time, at 33 times its budget, so most of its detections are events inside standing alarms; the tracker reached the same recall in alarm for 19% of the unlabelled time, at 6.5 times the budget. On the 13 h of sealed attack-free rows the frozen threshold ran at 0.92 times its budget, so those alarms belong to the attack recording, and we cannot tell disturbance that the attacks left outside their labels from false alarms. Here adaptation cost no recall and cut the alarm time outside the labels; tighten-only equalled the frozen threshold (33 times the budget) and the tracker under a ceiling of $1.25\,\tau_0$ ran at 25. P2 and P3 held: the capped tracker and every ceiling rule detected the same 25 events. With D2 (secondary) the tracker lost 9 of the 25 events ($[-0.44, -0.12]$). On the sealed rows (six overlapping placements of a 6 h campaign, descriptive) the loud pulses left the tracker with none of 6 bumps of $1.05\,\tau_0$ and the sub-event pulses left the hold-off tracker with none of 6, but the frog did not move SPOT, whose level rose by 0.6% and which caught all 6 (P4 failed). A code audit after the final found that the loader leaves valid the 36 steps after a recording gap, whose scores still depend on filled values. Rescored with those steps masked (exploratory; the recording has three gaps), frozen and tracker each detected 23 events and the frozen threshold's alarm time was unchanged.

# 7 Learning regions under attack (exploratory)

The analyses in this section were not registered. They use the spent HAI releases for development and reopen the three tt1 datasets after their finals; every access is in the ledger.

## 7.1 Extreme-value thresholds escape natural poisoning on the HAI data

On the three spent releases the moments SPOT, the maximum-likelihood SPOT and DSPOT stayed within two events of the frozen threshold on each release (recall 0.58 to 1.00 against 0.60 to 1.00), where the tracker lost 22 to 37 points; post-hoc on HAI 20.07 they caught 36, 35 and 36 of 38 attacks against 36 (Table 16 in Appendix C).

The exceptions: on HyGrid-ICS the moments SPOT lost 9 points ($[-0.15, -0.03]$); on voraus-AD, 190 calibration samples leave about four tail points above SPOT's base level (the calibration 98th percentile), and SPOT ran at 32% false positives against a 5% target. On the HAI data the extreme-value thresholds started near calibration: at the start of both HAI 20.07 test files the moments SPOT stood at $1.09\,\tau_0$, DSPOT at 1.03 to 1.05 and MLE SPOT

**Table 11: Tennessee Eastman, any-alarm count: events detected, then the mean and central 95% range of the count under random circular shifts of the rule's alarms. Data status: exploratory (tt1f).**

| Rule | first (141) | DoS×3 first (30) | DoS 10 h (41) | noise first (36) | DoS followers (45) | noise followers (59) |
|---|---|---|---|---|---|---|
| frozen | 116; 60.6 [51, 69] | 18; 13.0 [8, 17] | 28; 18.1 [12, 24] | 36; 29.2 [26, 33] | 45; 20.1 [14, 26] | 59; 40.3 [34, 47] |
| tracker | 116; 56.5 [48, 66] | 18; 10.8 [6, 15] | 28; 21.0 [14, 27] | 36; 24.5 [20, 29] | 45; 16.3 [10, 23] | 21; 29.2 [21, 36] |
| daily | 106; 53.9 [45, 63] | 13; 10.8 [6, 15] | 23; 17.9 [12, 25] | 36; 24.9 [20, 29] | 44; 17.0 [12, 23] | 15; 29.9 [23, 38] |
| daily, episodes left out | 113; 65.8 [56, 75] | 19; 13.8 [9, 19] | 24; 21.7 [14, 28] | 36; 30.0 [26, 33] | 45; 21.2 [15, 28] | 59; 41.1 [34, 48] |
| tighten-only | 122; 69.3 [60, 78] | 20; 14.2 [8, 19] | 32; 24.9 [19, 31] | 36; 29.9 [27, 33] | 45; 21.9 [16, 29] | 59; 41.7 [35, 48] |
| tracker, capped rise | 130; 84.8 [78, 91] | 26; 19.1 [14, 23] | 34; 33.1 [28, 37] | 36; 32.3 [30, 35] | 45; 29.9 [24, 35] | 59; 44.6 [39, 51] |

**Table 12: Matched operating points on Tennessee Eastman. For each rule, one multiplier $c$ puts the static threshold $c\,\tau_0$ at the rule's number of false-alarm episodes on the 56 sealed runs (an in-sample match). Detections on the attack runs: first events, then followers under the any-alarm / onset counts. Difference: recall of the static threshold minus recall of the rule on first events, runs resampled. False alarms: episodes over the budget on the unlabelled time of the attack runs. Data status: exploratory (tt1f).**

| Rule | $c$ | Rule | Static at its rate | Difference, first events | FA, rule / static |
|---|---|---|---|---|---|
| tracker | 0.985 | 116, 66 / 37 | 119, 104 / 44 | +0.021 $[-0.028, 0.071]$ | 0.82 / 1.77 |
| tracker, ceiling $1.25\,\tau_0$ | 0.979 | 121, 104 / 41 | 121, 104 / 44 | +0.000 $[-0.050, 0.050]$ | 1.26 / 1.89 |
| daily, episodes left out | 0.978 | 113, 104 / 44 | 121, 104 / 44 | +0.057 $[0.014, 0.106]$ | 1.87 / 1.91 |
| tighten-only | 0.965 | 122, 104 / 44 | 127, 104 / 46 | +0.035 $[0.007, 0.071]$ | 1.81 / 2.20 |
| tracker, capped rise | 0.942 | 130, 104 / 44 | 130, 104 / 46 | +0.000 $[-0.050, 0.050]$ | 2.35 / 2.80 |

**Table 13: Campaigns on the 56 sealed attack-free Tennessee Eastman runs: terminal bumps of 1.05 / 1.25 / 1.5 / 2 $\tau_0$ caught after the campaign, of 56. Without a campaign the adaptive rules caught 44 to 51 bumps of $1.05\,\tau_0$ and 55 or 56 of the larger ones; the frozen threshold caught all 56. Data status: registered (tt1d).**

| Rule | 24 h frog | Loud pulses | Sub-event pulses |
|---|---|---|---|
| frozen | 56 / 56 / 56 / 56 | 56 / 56 / 56 / 56 | 56 / 56 / 56 / 56 |
| tracker (as registered in tt1) | – | 0 / 0 / 0 / 56 | 0 / 0 / 0 / 0 |
| daily recalibration | – | 0 / 0 / 0 / 0 | 0 / 0 / 0 / 19 |
| daily, alarm episodes left out | 47 / 56 / 56 / 56 | 49 / 56 / 56 / 56 | 0 / 0 / 0 / 29 |
| tracker, rise stopped after 60 s of alarm | – | 0 / 0 / 0 / 56 | 0 / 0 / 0 / 0 |
| tracker, rise capped at $0.1\,\tau_0$ per day | 51 / 56 / 56 / 56 | 8 / 55 / 56 / 56 | 6 / 55 / 56 / 56 |
| SPOT, maximum likelihood | 0 / 16 / 24 / 32 | 44 / 56 / 56 / 56 | 44 / 56 / 56 / 56 |
| SPOT, ceiling $1.25\,\tau_0$ | 0 / 16 / 56 / 56 | – | – |
| tracker, ceiling $1.25\,\tau_0$ | 46 / 56 / 56 / 56 | 0 / 56 / 56 / 56 | 0 / 56 / 56 / 56 |
| daily, ceiling $1.25\,\tau_0$ | – | 0 / 0 / 56 / 56 | 0 / 0 / 56 / 56 |
| SPOT, approval every 6 h / daily | 1 / 17 / 24 / 40, 32 / 42 / 53 / 56 | – | – |

at 1.18 (post-hoc). Section 6.1 shows what a heavier calibration tail does to that starting point.

## 7.2 Other scores

With D0 the tracker lost recall on every dataset where it was measured: $[-0.38, -0.14]$, $[-0.21, -0.05]$ and $[-0.19, -0.02]$ on the spent releases (exploratory), $[-0.32, -0.08]$ on HAI 20.07, $[-0.12, -0.02]$ on HyGrid-ICS and $[-0.04, -0.01]$ on voraus-AD (post-hoc), and $[-0.86, -0.36]$ on BATADAL (registered tt1b). With the reconstruction score D2 (exploratory here), the frozen threshold caught 65 of the 160 events of the development releases and the tracker 53 ($[-0.14, -0.01]$) pooled; on HAI 22.04 the tracker tightened instead and caught 18 against 11, at 31 false alarms a day. The registered D2 results differ by dataset: no loss on Tennessee Eastman, where the tracker also tightened (Section 6.2), and a loss on ProvICS (Section 6.4).

## 7.3 Mechanism: attacks that follow attacks

The tracker's misses concentrate on attacks that start within 6 h of the previous attack's end. D1 recall on close versus isolated attacks was 0.75 / 1.00, 0.41 / 1.00 and 0.20 / 0.67 on the spent releases and 0.48 / 0.71 on HAI 20.07 (31 close, 7 isolated). The frozen threshold also detected close attacks less often on two releases, so part of the gap reflects harder attacks; the tracker's gap is larger on every release. On these releases the onset count changes almost nothing: it credits the frozen threshold with 124 of 160 events and the tracker

with 76 ($-0.30$, $[-0.37, -0.23]$), against 125 and 77 under the any-alarm count.

Attacks are the main thing that raises the tracker. Rerun post-hoc with its update fed only the attack-labelled samples (the others replaced by the calibration median for the update, alarms still on the real score), the tracker detected 20 of 38 on HAI 20.07 and 14 of 91 on HyGrid-ICS, exactly its registered counts, and 7 of 14 on BATADAL (6 registered). Fed only the unlabelled samples, it detected 28 of 38, 27 of 91 (equal to frozen) and 11 of 14: the unlabelled exceedances add a smaller loss of their own, $-0.21$ $[-0.34, -0.08]$ on HAI 20.07 and $-0.21$ $[-0.43, 0.00]$ on BATADAL.

## 7.4 Campaigns and stronger rules in development

The campaigns and the stronger rules that Section 6 registers were developed on normal data of the spent releases; Appendix C gives the experiments and their tables. In short: twelve supra-threshold setpoint pulses a day left daily recalibration and both trackers catching 1 to 4 of 30 just-detectable terminal steps (controls 21 to 25); a score-level frog blinded every SPOT and DSPOT variant, and a version that spoofs noise onto three sensor readings blinded the maximum-likelihood SPOT in 8 of 12 placements; recalibration that leaves alarm episodes out and a tracker with a capped rise avoided natural poisoning, and each was then pushed up through what it does not exclude; a ceiling confined every campaign to bumps at or below it; and an approval gate that reads the alarm log approved the silent campaign.

## 7.5 Sensor-level campaigns (exploratory)

The registered campaigns write the score. In a last exploratory round the attacker writes three sensor readings, those most coupled to one setpoint channel, and the real monitor scores the result on normal data of the spent releases (30 placements, Table 23 in Appendix F); the terminal attack is a setpoint step of one to three times the smallest step the frozen threshold detects, and the plant does not respond to the spoofed readings. A frog that reads the threshold and sizes its noise on the monitor left the standard SPOT with 9 of 30 smallest steps and 21 at 1.5 times that size (controls 30 and 30), and raised no alarm of the attacked rule in 12 placements. An attacker who sees only alarms, raising its noise until one fires and halving it afterwards, spent a median 13 alarm episodes in six hours and left SPOT with 20 of 30 smallest steps. Sensor pulses of 20 s, sized so that the alarm stays shorter than an event, left the hold-off tracker with 0 of 30 smallest steps and 1 at three times that size (controls 21 and 30) and cost the excluding recalibration 9 of 30 smallest steps. None of these campaigns beats replay. An attacker who shows clean values on the same three sensors during the terminal step hides it from the frozen threshold in 27 of 30 placements at the smallest size and in 19 at three times that size. With this access, poisoning the threshold adds nothing to concealing the attack directly.

# 8 Design guidance

A threshold rule's learning region is its attack surface. Every automatic loosening path we put through a campaign could be pushed upward at the score level: exceedance tracking and daily recalibration by loud pulses, tail refitting by a closed-loop attacker, and exclusion rules by pulses below their event length (registered on Tennessee Eastman). Recorded attacks raised the plain tracker and plain daily recalibration by themselves (registered), with the qualifications of Sections 5.2 and 6; rules that leave alarm episodes out or cap the rise were not raised by recorded attacks. The recommendations are ordered by how well they are supported.

(1) **Keep a fixed reference threshold.** A frozen $\tau_0$ caught every terminal attack of every campaign, and a $\tau_0$ monitor beside the attacked rule would have been in alarm for a median 99% of the silent campaign. If the deployed rule adapts, alarm on the fixed threshold as well, or log $\tau_t$ against $\tau_0$: a threshold that rises during an incident, or climbs without alarms, is itself an indicator.
(2) **If a threshold must loosen by itself, give it a ceiling.** Under a ceiling $\kappa\tau_0$ no poisoning can hide an event above $\kappa\tau_0$ (Proposition 4), and nothing more is guaranteed: on Tennessee Eastman the static threshold $1.1\,\tau_0$ caught 80 of the 141 first attacks against 116 for $\tau_0$, and 70 of the 141 never exceed $1.25\,\tau_0$. Scale an adaptive step by $\tau_0$, not by a calibration spread: on BATADAL the spread-scaled tracker lost 8 of 14 attacks, and under a ceiling it lost none and ran at 36 to 43 times the budget, where the $\tau_0$-scaled one lost none at 0.4 to 0.6 times (exploratory).
(3) **Do not expect detection from automatic tightening.** A tracker capped at the calibrated threshold cannot lose an event the frozen threshold catches under the any-alarm count (Proposition 3). It ran at 1.57 times the false-alarm budget on the sealed Tennessee Eastman runs against 0.98 for frozen, and at 2.43 against 0.70 on BATADAL (registered), and a frozen threshold lowered by 3.5% to the same alarm rate detected more first attacks (127 against 122, exploratory).
(4) **Loosening beyond the ceiling, and recalibration, belong to change control** (not tested here). Process-alarm standards route alarm changes through authorised management of change [11, 20], and recalibration on verified clean data is the established remedy when drift raises false alarms [8]. Our only evidence is negative: a gate that approved loosening whenever the alarm log was quiet let the silent campaign through, and an oracle that blocked loosening only during labelled events caught 1 of 10 terminal steps. We tested no human operator and no recalibration procedure.
(5) **Budget alarm time as well as episodes.** On HAI 20.07 a 24-hour alarm that overlapped seven attacks did not count against the episode budget, and an attack that begins inside a running alarm gives the operator nothing new to see.

The frozen threshold is the reference in all of this, and it did not always hold its budget. It ran at 0.98 times the budget on the sealed Tennessee Eastman runs, 0.70 on BATADAL and 0.92 on 13 h of sealed ProvICS rows, but at 1.55 on the Tennessee Eastman attack runs, 1.11 on HAI 20.07 with 39% of the unlabelled time in alarm, and 33 on the ProvICS attack recording, where the plain tracker ran at 6.5 with the same detections, tighten-only inherited

the whole cost and a ceiling of $1.25\,\tau_0$ most of it (25). No dataset here has labelled benign drift, so the case adaptive rules exist for, a plant that changes without being attacked, is the one this paper cannot evaluate. Earlier work proposes lowering the threshold against poisoning of retrained ICS detectors [23] and shows that a bounded false-positive rate limits a poisoner [22]; what we add is the measurement of each loosening path and of the ceiling.

## 9 Limitations

- Three of the four registered primary rules failed: tt1b's through H3, and both of tt1d's, through the onset count on Tennessee Eastman and through P1 on ProvICS. The registered support is the individual hypotheses that passed, under the any-alarm count, and no multiplicity correction was registered. Under the onset count the tracker's loss survives on BATADAL only among the registered datasets (post-hoc, and only with the registered step) and on the development releases (exploratory).
- The any-alarm count credits an attack that falls inside an alarm already running and, on long attacks, an alarm that would have come anyway. The chance references of Sections 5.2 and 6.2 are exploratory, and H1c is weak evidence for that reason. We do not model operator response time.
- On Tennessee Eastman the tracker's whole follower loss is in sequences of noise attacks with no gap between attacks, each of which it had flagged at its start. Whether that loss matters depends on what an operator does with a day-long alarm, which we do not model.
- The registered campaigns are score-level injections on one simulated process whose sealed runs come from the simulator and modes of the calibration runs; the 56 placements are seven monitor-level outcomes. On ProvICS the frog did not move SPOT. The sensor-level campaigns on HAI are exploratory and replay recorded data: the plant does not respond to the spoofed readings.
- HAI 20.07 shares a testbed with the development releases, and its monitor drifted (39% of the unlabelled time in alarm). HyGrid-ICS is a single simulated day; voraus-AD anomalies are not attacks. No dataset has labelled benign drift.
- The intervals on HAI 20.07 and BATADAL resample events as if they were independent, although events in one stream share the threshold's state; they are probably too narrow.
- The registrations are commits in our own repository, made seconds before the access they govern (Section 4), with no external timestamp. A code audit after the finals found five defects in the frozen analysis code (Appendix F). Four touch code behind registered numbers. Two could move one materially: the calibration search, under which the D1 thresholds of Tennessee Eastman and of the HAI development releases are unchanged (HAI 20.07, BATADAL and ProvICS were not rechecked), and the ProvICS validity mask (Section 6.4). The other two, in the first smoothing window and in the hold-off tracker's episode counter, were not rechecked.
- Our monitor is one world-model residual, one model-free score and a linear reconstruction score; nonlinear detectors were not tested. SPOT's behaviour depends on the tail estimator and on the score's tail shape. We tested no human operator; human-approved suppression of detection rules has been observed to accumulate and persist [9].
- The ProvICS result is registered with one dated amendment after a failed first attempt. No claim of runtime authority, deployment safety or incident prevention follows from these results.

## 10 Conclusion

A quantile tracker that learns from exceedances alarms on a sustained attack, climbs through it and goes quiet while the attack continues. Under a count that credits any alarm inside an attack this cost detections on HAI 20.07, BATADAL and Tennessee Eastman (registered). Follow-up analysis narrows each result: the Tennessee Eastman follower losses are later attacks of noise sequences the tracker had flagged, the BATADAL loss needs a step scaled by a heavy-tailed spread, part of the frozen threshold's lead on HAI 20.07 is standing alarm, and ProvICS shows no loss. One of four registered primary rules passed. How long a tracker stays raised follows from its calibrated false-alarm rate alone, so making it slower does not make it forget faster.

Deliberate poisoning, injected at the score level, was confirmed on sealed attack-free runs of one simulated process: loud pulses blinded the tracker, sub-event pulses blinded the recalibration that leaves alarms out, and a closed-loop attacker blinded a standard SPOT in five of seven operating modes, while the frozen threshold caught every later attack. That silent campaign is silent only to its victim; a fixed threshold beside it sees a day-long alarm. A ceiling confines what any of these attacks can hide to events at or below it. Automatic tightening kept the frozen threshold's detections by construction, cost up to 3.5 times its false alarms, and detected no more than a slightly lower frozen threshold. An adaptive alarm threshold should therefore run under a ceiling and beside a fixed reference; whether adapting is worth it at all is a question for data with benign drift, which this paper does not have.

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## Ethical considerations

All experiments ran offline on public recordings: HAI 20.07 to 23.05, HyGrid-ICS, voraus-AD, the BATADAL re-simulation, the Tennessee Eastman attack set and ProvICS, each used under its publisher's terms. No live plant, testbed, vendor product or operator was involved, and no human subjects took part. The pulse campaigns and both frog attacks were injected into recorded normal data or into the score; the physically realizable frog changes recorded sensor values in software and is scored by our own monitor.

The attacks are not new in kind. Boiling-frog poisoning of self-updating detectors has been public since 2009 [6, 26], and the threshold rules we attack are textbook algorithms. What the paper adds for an attacker is the observation that an adaptive threshold widens the window for a later attack; what it adds for a defender is a rule that closes that window at a known false-alarm cost. We judged that publishing both helps defenders more than attackers, because the defence is a configuration change (cap the threshold at its calibrated value) that any operator of an adaptive threshold can apply today. The released code generates attacks only against recorded data and a score file; it contains no tooling for reaching a control network.

**Table 14: Tighten-only with a loosening permit on HAI 20.07. Data status: registered (tt1).**

| Permit | Recall | vs frozen (95%) | FA/day | Permits/wk |
|---|---|---|---|---|
| spread, daily review | 26/38 | $[-0.39, -0.13]$ | 1.62 | 2.7 |
| spread, 6 h review | 28/38 | $[-0.34, -0.08]$ | 2.63 | 4.1 |
| shift, daily review | 32/38 | $[-0.21, -0.03]$ | 1.01 | 4.1 |
| shift, 6 h review | 31/38 | $[-0.24, -0.03]$ | 2.02 | 12.2 |
| oracle | 36/38 | $[0, 0]$ | 2.23 | 0 |

## Open science

The three pre-registrations, the data-access ledger, the attack and threshold harnesses, the registered-gate verifiers and the figure data and plotting code are available in the public repository at https://github.com/VyomKulshrestha/Ferrum-OS (paper sources, results and verifier reports under `docs/research`), which also contains the result files behind every number in this paper and the verifier reports (27 of 27 checks for tt1, 8 of 8 for tt1b, 19 of 19 and 22 of 22 for tt1d). The raw datasets are not redistributed here and are available from their publishers: HAI from its dataset repository, BATADAL from Taormina et al. [33] and the no-concealment re-simulation from the artifact of Erba et al. [12], voraus-AD from Brockmann et al. [4], HyGrid-ICS from its Zenodo record [37], the Tennessee Eastman attack set from Kaspersky [13] and ProvICS from its authors [28], under whose terms it is not redistributed. Reproducing the registered results requires one access to each raw dataset under the access rules recorded in the ledger. The repository's commit history shows that every registration was committed before the data access it governs.

## A Other registered outcomes of tt1

Table 14 and the list below complete the registered record of tt1.

- **Automatic permits.** Every automatic loosening permit lost recall on HAI 20.07. On voraus-AD all four lost recall too (95% intervals: spread $[-0.21, -0.15]$ with daily review and $[-0.28, -0.21]$ with 6 h review; shift $[-0.18, -0.13]$ and $[-0.21, -0.15]$); on HyGrid-ICS only the 6 h shift permit did (24 of 91, $[-0.08, 0.00]$).
- **Oracle permit.** On the real attacks it equalled frozen, but in the G1 campaign runs it caught the terminal step in 7 of 10 placements after the pulses and 1 of 10 in the control.
- **First and second half of the HAI 20.07 events.** Recall was 0.947 / 0.947 for frozen, 0.474 / 0.579 for the tracker, 0.579 / 0.842 for the slow tracker and 0.474 / 0.474 for daily recalibration.
- **E5, sub-threshold noise campaign.** Only 1 of 13 placements admitted a stealthy noise level; 12 had none, so E5 gives no result.
- **BATADAL frog variants (tt1b).** The registered frog variants (10 placements each) did test MLE SPOT. Under the 24 h closed-loop frog it caught the terminal attack in 7 of 10 placements against 9 of 10 in its control with D1, and in 0 of 10 against 4 of 10 with D0; with 6 h and 1 h campaigns it matched its D1 control and caught 2 and 4 of 10 against 4 with D0. Frozen and tighten-only caught 10 of 10 in every variant, daily recalibration 9 of 10, the margin none, and the tracker, SPOT and DSPOT at most 1 of 10 with or without a campaign. The frozen threshold detected the 14 real attacks a median 3.5 steps (52.5 minutes) after onset and at most 65 steps (16.3 h) after.
- A separate registration that shared the HAI 20.07 and HyGrid-ICS access (detection-v2) is reported on its own.
- **A superseded post-hoc variant.** Before the onset count of Section 5.2 we hid each HAI 20.07 file's opening run above $\tau_0$ from the threshold update, which gave tracker $-0.29$ $[-0.45, -0.16]$ and slow tracker $-0.08$ $[-0.18, 0.00]$. We superseded it because the opening run of test1 turned out to contain seven attacks.

**Table 15: The physically realizable frog. Data status: exploratory (normal data of the spent releases).**

| Rule | Terminal steps caught, after campaign / in control (of 12 placements) | Threshold at the terminal step (median per release) |
|---|---|---|
| frozen | 12 / 12 | 1.0 $\tau_0$ |
| SPOT (MLE) | 4 / 12 | 1.04 to 1.59 $\tau_0$ |
| SPOT (moments) | 6 / 12 | 0.78 to 1.42 $\tau_0$ |
| DSPOT | 9 / 9 | 0.87 to 1.12 $\tau_0$ |

## B The physically realizable frog

The score-level attacker sets the score directly. We also gave the attacker only the readings of three sensors coupled to one setpoint channel. Every 5 minutes it reads the rule's threshold and spoofs the largest zero-mean noise whose D1 score, computed by the real world model on the spoofed history, stays at 0.97 of that threshold; after 6 h it stops, and 1 h later the just-detectable setpoint step follows. This attacker is white-box: it can run the monitor.

Controls caught 12 of 12 for frozen and both SPOT variants and 9 of 12 for DSPOT, so the campaign did not change DSPOT's total (per release, campaign against control: 4 against 3, 3 against 4 and 2 against 2). On HAI 22.04 both SPOT variants missed the terminal step on all four placements, and the median campaign raised no alarm. On HAI 23.05 it was not silent: the frozen threshold raised a median of 190 alarm steps during the campaign, and the SPOT variants 154 to 171. The spoofed noise had a scale of 0.02 to 0.04 robust units against frozen and the SPOT variants, and 0.05 to 0.12 against DSPOT, whose moving mean follows the noise-driven rise, so the attacker's cap rises with the mean rather than with the tail.

## C Development experiments in detail

Tables 16 to 19 and Figure 3 give the per-rule results of the exploratory experiments of Section 7. Figure 4 collects the recall differences of tt1, tt1b and the development releases with their status.

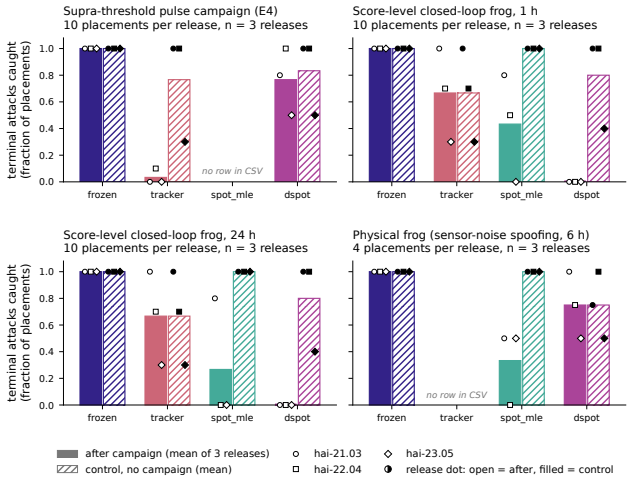



**Figure 3: Terminal attacks caught after each campaign (solid) and in the matching control (hatched), for frozen, tracker, spot_mle and dspot. Each bar is the mean over the three spent HAI releases; dots are the per-release values, and every release had the same number of placements, so the mean equals the pooled fraction. Panels: the supra-threshold pulse campaign (E4, 10 placements per release); the score-level closed-loop frog at 1 h and 24 h (10 placements); and the physical frog, which spoofs noise on three sensors for 6 h (4 placements). Data status: all four panels are exploratory results on the normal data of the spent releases. The source CSV has no spot_mle row for E4 and no tracker row for the physical frog; both slots are left empty and labelled.**

One experiment of the second plan is not in Table 20: the pulse campaign of Section C.1 against the ceiling rules. Under ceilings of 1.1, 1.25 and 1.5 $\tau_0$ the tracker caught 20, 9 and 3 of 30 just-detectable terminal steps (controls 28, 24 and 24) and daily recalibration 17, 11 and 6 (controls 24, 23 and 22), no fewer than the static thresholds at those levels (15, 9 and 3); the standard SPOT, its ceilings and both approval gates caught all 30.

## C.1 Loud poisoning: pulse campaigns

Normal data of the spent releases, 30 placements; the terminal step is the smallest setpoint step the frozen threshold detects on the clean stream. After a campaign of 120 s supra-threshold pulses, twelve a day for up to a day, daily recalibration caught 4 of 30 terminal steps (control 21), the tracker 1 (23) and the slow tracker 3 (25); the frozen threshold caught all 30 and SPOT 29 (Table 17 and Figure 3 in Appendix C).

Open-loop sub-alarm pulses did not poison SPOT: its threshold fell (0.82 to 0.98 $\tau_0$), and many pulses crossed SPOT's own alarm level and were refused.

## C.2 Silent poisoning: the closed-loop frog

Score-level attacker A2, normal data of the spent releases, 10 placements each: the attacker holds the score at 0.97 $\tau_t$, then, after a one-hour lead, a 600 s bump at 1.05 $\tau_0$ follows (600 s is the terminal length of every development campaign). After 24 h the moments SPOT and DSPOT caught none of 30 terminal bumps and the maximum-likelihood SPOT 8; the frozen threshold, the slow tracker and tighten-only were unaffected (Table 18).

Each rule is compared with its own control (the same bump without a campaign). The frozen and both SPOT controls caught 30/30 and DSPOT's 24/30; the tracker and daily controls already missed some bumps on the drifting releases, and the campaign did not lower them further. After 1 h the moments SPOT threshold stood at 1.1 to 4.5 $\tau_0$; after 24 h it had grown by more than five orders of magnitude, and the MLE fit left the floating-point range on one release. A synthetic control with a light tail (the 60-step moving average of $|N(0,1)|$, five seeds) did not run away: the moments SPOT threshold stayed at 0.95 to 0.97 $\tau_0$ and caught every terminal bump, and maximum-likelihood SPOT rose to a median of 1.02 to 1.04 $\tau_0$ (at most 1.06), though it missed the bump on one of the five seeds in the 24 h, 6 h and fixed-level variants. The runaway therefore depends on the heavy tail of real residual scores.

*A physically realizable version.* We also gave the attacker only the readings of three sensors coupled to one setpoint channel, spoofed with zero-mean noise that keeps the D1 score at 0.97 of the rule's threshold for 6 h (Appendix B). The maximum-likelihood SPOT caught 4 of 12 terminal steps against 12 in its control and the moments version 6; DSPOT was unchanged (9 against 9). On HAI 22.04 the median campaign raised no alarm; on HAI 23.05 it was not silent.

## C.3 Stronger baselines and a standard SPOT (exploratory)

The rules above are the plain versions. After the mock review we fixed four stronger adaptive rules and three SPOT variants in a written plan and ran them on the spent development releases (the plan was committed before the runs; one rule was corrected by a dated amendment after the first run, and every run was then repeated). They are daily recalibration that leaves out alarm episodes longer than 60 s and the hour after them; the tracker with its upward steps stopped once an alarm has lasted 60 s and for an hour after; a tracker whose rise is capped at 0.1 $\tau_0$ per day; Telemanom's dynamic threshold [19]; SPOT with a maximum-likelihood tail refitted as the peaks grow by 1%, also with the shape bounded to $|\xi| \leq 0.5$ and with runs declustering; and DSPOT on the maximum-likelihood tail (Table 19 in Appendix C).

Leaving events out of the update, or capping the rise, removed natural poisoning on these releases: the excluding rule matched frozen recall and the capped tracker missed one more event. Stopping the tracker after a minute of alarm did not, because by Proposition 1 the first minute of each attack still keeps it raised for about a day. The excluding rule also resisted the pulse campaign, whose 120 s pulses count as events. An attacker who knows the 60 s event length can keep each exceedance shorter; Section C.4 runs that attack. The capped tracker lost a little to the pulses, and its cap bounds how fast it rises, not how far. Telemanom's threshold caught more attacks at 16 to 19 times the false-alarm budget, so its recall is not comparable with the others.

**Table 16: Event recall of the extreme-value rules. Data status: exploratory on the spent releases, post-hoc on HAI 20.07.**

| Data | frozen | tracker | SPOT (moments) | SPOT (MLE) | DSPOT |
|---|---|---|---|---|---|
| HAI 21.03 / 22.04 / 23.05 | 1.00 / 0.76 / 0.60 | 0.78 / 0.45 / 0.23 | 1.00 / 0.79 / 0.58 | 1.00 / 0.79 / 0.60 | 1.00 / 0.79 / 0.60 |
| HAI 20.07 (post-hoc) | 0.947 | 0.526 | 0.947 | 0.921 $[-0.08, 0.00]$ | 0.947 |

**Table 17: Terminal attacks caught after a pulse campaign (control in parentheses). Data status: exploratory (normal data of the spent releases).**

| Campaign | frozen | daily | tracker | tracker_slow | SPOT | DSPOT |
|---|---|---|---|---|---|---|
| 12 × 120 s supra-threshold pulses per day | 30/30 (30/30) | 4/30 (21/30) | 1/30 (23/30) | 3/30 (25/30) | 29/30 (30/30) | 23/30 (25/30) |
| 24 × 300 s sub-alarm pulses per day, open loop | 30/30 | 23/30 (23/30) | 19/30 (21/30) | 23/30 | 29/30 | 22/30 (24/30) |

**Table 18: Terminal attacks caught after a closed-loop frog campaign, 30 placements. Data status: exploratory (normal data of the spent releases).**

| Rule | 24 h campaign | 6 h | 1 h | open loop, fixed level, 24 h |
|---|---|---|---|---|
| frozen, tracker_slow, tighten-only | as control | as control | as control | as control |
| tracker | as control | as control | as control | 16/30 (control 20/30) |
| SPOT (moments) | 0/30 | 0/30 | 0/30 | not moved; alarms throughout |
| DSPOT | 0/30 | 0/30 | 0/30 | not moved; alarms throughout |
| SPOT (MLE, refit every 2048 peaks) | 8/30 | 8/30 | 13/30 | 11/30 |

The standard SPOT did not escape the closed-loop attacker. Refitted by maximum likelihood, with or without a bounded shape, it missed the terminal step in all 30 placements of the 24 h campaign, and DSPOT on the same tail in all 30. Declustering resisted a continuous frog, which forms a single cluster, but an attacker who leaves the score clean for two minutes after each minute of pushing blinded it on HAI 22.04 (10 of 30 pooled, against a control of 20).

## C.4 Bounded loosening, sub-event pulses and an approval gate (exploratory)

A second written plan, committed before its runs, tested three things on the same releases. *A ceiling*: the tracker, daily recalibration and the standard SPOT with the deployed threshold clipped at $\kappa\tau_0$ (Proposition 4). *Sub-event pulses*: a score-level attacker who knows the 60 s event length sets the score to twice the current threshold for 45 s every 30 minutes for a day, so that a pulse by itself is too short to count as an event. *An approval gate*: SPOT proposes its level, and a ceiling rises to it at each review if the review window held no alarm, a stand-in for an operator who signs a loosening when the alarm log is quiet. Campaigns end in terminal bumps of 1.05 to 10 $\tau_0$ (Table 20).

With a ceiling at 1.1 $\tau_0$ the tracker and daily recalibration caught the 125 events the frozen threshold caught. Wider ceilings cost recall, as Proposition 4 allows: at 1.25 and 1.5 $\tau_0$ the tracker caught 118 and 108, against 116 and 107 for the static thresholds at those levels. The frog shows what a ceiling buys. Without one, the standard SPOT missed every terminal bump up to 1.5 $\tau_0$, and on two of the three releases every bump up to 10 $\tau_0$, because its threshold left the floating-point range. With a ceiling it missed only bumps at or below the ceiling and caught all 30 above it.

The sub-event pulses defeated both rules that depend on recognising an event. The excluding recalibration caught none of 30 terminal bumps up to 1.25 $\tau_0$ (control 30) and the tracker with a hold-off none at any size up to 10 $\tau_0$, since the pulses were too short to be left out. The capped tracker missed 17 of 30 at 1.05 $\tau_0$ and none from 1.25 $\tau_0$. SPOT, which does not learn from alarms, was unaffected. The quiet-log gate did not stop the frog: with a review every 6 h it caught 15 of 30 bumps at 1.05 $\tau_0$ and 20 of 30 at 2 $\tau_0$; with a daily review, 25 of 30 at 1.05 $\tau_0$ and all 30 above. A campaign that raises no alarm passes an approval that reads the alarm log.

**Table 19: Stronger adaptive rules and a standard SPOT on the three spent HAI releases. Natural poisoning: events detected of 160 and the paired difference against frozen (125 of 160). Pulses and frog: terminal steps caught after the campaign / in the control, of 30 placements. Data status: exploratory.**

| Rule | Natural poisoning | Pulses | 24 h frog |
|---|---|---|---|
| tracker (as registered) | 77, $[-0.37, -0.23]$ | 1 / 23 | as control |
| daily recalibration | 104, $[-0.19, -0.08]$ | 4 / 21 | as control |
| daily, alarm episodes left out | 125, $[-0.02, 0.02]$ | 30 / 30 | 30 / 30 |
| tracker, rise stopped after 60 s of alarm | 89, $[-0.29, -0.16]$ | 2 / 24 | 21 / 21 |
| tracker, rise capped at $0.1\,\tau_0$ per day | 124, $[-0.02, 0.00]$ | 25 / 30 | 30 / 30 |
| Telemanom dynamic threshold (33 to 37 false alarms a day) | 140, $[+0.04, +0.15]$ | 27 / 26 | 28 / 28 |
| SPOT, maximum likelihood | 126, $[-0.01, +0.03]$ | 30 / 30 | 0 / 30 |
| SPOT, maximum likelihood, $\lvert\xi\rvert \leq 0.5$ | 127, $[0.00, +0.03]$ | – | 0 / 30 |
| SPOT, maximum likelihood, declustered | 123, $[-0.04, +0.02]$ | 21 / 24 | 20 / 20 (bursts: 10 / 20) |
| DSPOT, maximum-likelihood tail | 127, $[0.00, +0.03]$ | – | 0 / 24 |

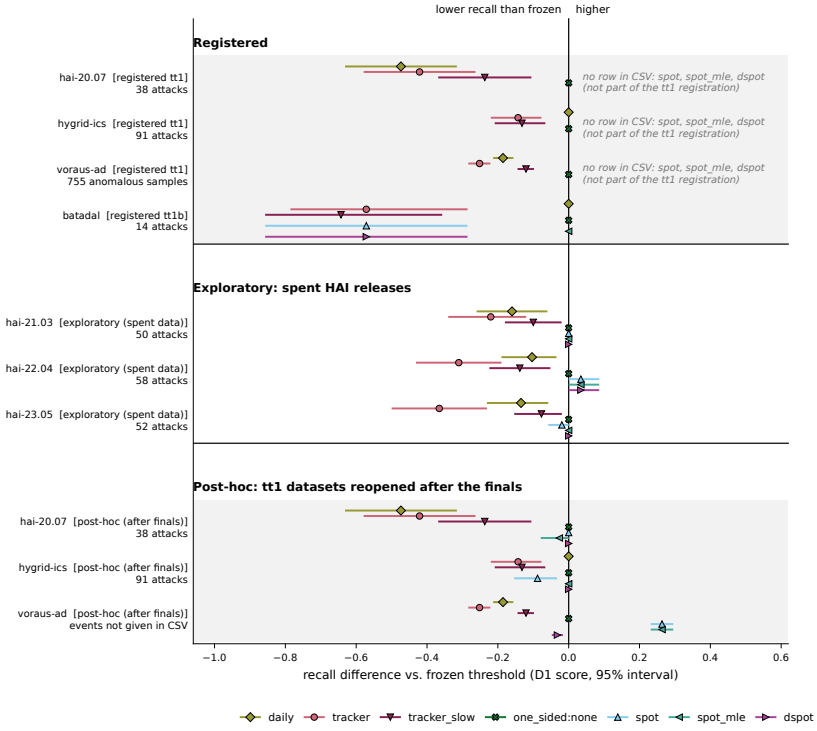



**Figure 4: Paired difference in event recall between each adaptive rule and the frozen threshold on D1, with 95% intervals. Values left of 0 mean lower recall than frozen. Three blocks, each row labelled with its data status: (1) registered results from tt1 (hai-20.07, hygrid-ics, voraus-ad) and tt1b (batadal); (2) exploratory development results on the spent HAI releases; (3) post-hoc results on the tt1 datasets, reopened after the finals. spot, spot_mle and dspot have no rows for the registered tt1 datasets because those rules were not part of the tt1 registration; the gaps are labelled and the post-hoc block supplies the comparisons. one_sided:none equals frozen (0 $[0, 0]$) in every block. The positive spot and spot_mle values for voraus-ad (post-hoc) are not a gain: with 190 calibration samples SPOT ran at a per-sample false-positive rate of 0.32 against the 0.05 target (Section 7.1).**

**Table 20: Ceilings, sub-event pulses and an approval gate on the three spent HAI releases. Natural poisoning: events detected of 160. Campaigns: terminal bumps of 1.05 / 1.5 / 3 $\tau_0$ caught after the campaign, of 30 placements; where a rule's control at 1.05 $\tau_0$ was below 30 it is given in parentheses. Data status: exploratory.**

| Rule | Natural poisoning | 24 h frog | Sub-event pulses |
|---|---|---|---|
| frozen | 125 | 30 / 30 / 30 | 30 / 30 / 30 |
| tracker (as registered) | 77 | – | 0 / 0 / 0 (20) |
| tracker, ceiling 1.1 / 1.25 / 1.5 $\tau_0$ | 125 / 118 / 108 | – | 6 / 30 / 30 (25), 0 / 30 / 30 (20), 0 / 30 / 30 (20) |
| daily recalibration | 104 | – | 0 / 5 / 24 (18) |
| daily, ceiling 1.1 / 1.25 / 1.5 $\tau_0$ | 125 / 121 / 119 | – | 0 / 30 / 30 (18), 0 / 30 / 30 (18), 0 / 5 / 30 (18) |
| daily, alarm episodes left out | 125 | 30 / 30 / 30 | 0 / 11 / 30 |
| tracker, rise stopped after 60 s of alarm | 89 | as control | 0 / 0 / 0 (21) |
| tracker, rise capped at 0.1 $\tau_0$ per day | 124 | 30 / 30 / 30 | 13 / 30 / 30 |
| SPOT, maximum likelihood | 126 | 0 / 0 / 10 | 30 / 30 / 30 |
| SPOT, ceiling 1.1 / 1.25 / 1.5 $\tau_0$ | 126 / 126 / 126 | 0 / 30 / 30, 0 / 30 / 30, 0 / 7 / 30 | – |
| SPOT, approval every 6 h / daily | 127 / 127 | 15 / 19 / 30, 25 / 30 / 30 | – |

## D How the hypothesis changed during development

All development used normal training data and the already-opened attack files of HAI 21.03, 22.04 and 23.05. We report it because the registered claim differs from the one we set out to test.

(1) *Tighten-only with automatic loosening.* Our starting design let a threshold tighten itself and loosen only through a statistical permit. A slow sub-threshold drift ("boiling frog" in the style of Rubinstein et al.) did not poison any rule: the world model absorbed the drift. Supra-threshold pulses did inflate the trackers, but every automatic permit we tried either opened under the pulses (spread permit: 3 of 30 terminal attacks caught in development round r3, 5 of 30 in round r4) or failed to follow benign drift (shift permit).
(2) *Static safety margin.* A margin at half the calibration exceedance rate cut the false-alarm rate on pseudo-deployment splits without always reaching the budget (HAI 20.07 development: 3.79 episodes per day against 4.42 for frozen, budget 2), and cost 4 to 10 points of recall on the spent attacks (interval lower bounds down to −0.17), so we registered a smaller margin and made it secondary.
(3) *Frozen versus adaptive.* On the spent attacks the frozen threshold held the budget on all three real test periods (0.66, 1.48 and 1.27 episodes per day), while the tracker lost 22, 31 and 37 points of recall with intervals entirely below zero. This became the registered primary.

## E Every registered rule of tt1d

Table 21 gives the registered D1 outcomes of all twenty rules on both datasets. By attack type (21, 22, 23, 24; 34 / 75 / 41 / 95 events) the frozen threshold detected 34 / 63 / 28 / 95, the tracker 34 / 63 / 28 / 57, daily recalibration 34 / 57 / 23 / 51 and the capped tracker 34 / 71 / 34 / 95.

**Table 21: tt1d, D1, every registered rule. Tennessee Eastman: first events detected (of 141), followers detected under the any-alarm and onset counts (of 104), and false-alarm episodes over the budget on the unlabelled time of the attack runs and on the sealed attack-free runs. ProvICS: events detected under both counts (of 32) and false alarms over the budget on the unlabelled time of the attack recording. Data status: registered (tt1d; ProvICS under amendment 1).**

| | Tennessee Eastman | | | | | ProvICS | | |
|---|---|---|---|---|---|---|---|---|
| Rule | first | followers | onset | FA, attack runs | FA, sealed | any alarm | onset | FA |
| frozen | 116 | 104 | 44 | 1.55 | 0.98 | 25 | 4 | 33.0 |
| tracker | 116 | 66 | 37 | 0.82 | 1.21 | 25 | 4 | 6.5 |
| tracker_slow | 110 | 88 | 50 | 0.74 | 0.83 | 25 | 4 | 18.8 |
| tracker, $\eta = 0.01\,\tau_0$ | 123 | 55 | 37 | 1.02 | 1.64 | 24 | 6 | 1.9 |
| daily | 106 | 59 | 40 | 1.23 | 1.22 | 25 | 4 | 33.0 |
| tighten-only | 122 | 104 | 44 | 1.81 | 1.57 | 25 | 4 | 33.0 |
| daily, episodes left out | 113 | 104 | 44 | 1.87 | 1.33 | 25 | 4 | 33.0 |
| tracker, hold-off | 124 | 104 | 45 | 1.65 | 1.45 | 25 | 4 | 33.6 |
| tracker, capped rise | 130 | 104 | 44 | 2.35 | 2.12 | 25 | 4 | 37.5 |
| SPOT (ML) | 106 | 104 | 43 | 1.41 | 0.84 | 25 | 4 | 35.6 |
| static $1.1\,\tau_0$ | 80 | 104 | 45 | 0.83 | 0.32 | 25 | 4 | 37.5 |
| static $1.25\,\tau_0$ | 71 | 104 | 43 | 0.50 | 0.16 | 25 | 4 | 25.2 |
| static $1.5\,\tau_0$ | 71 | 104 | 44 | 0.25 | 0.11 | 25 | 4 | 5.8 |
| tracker, ceiling 1.1 | 123 | 104 | 43 | 1.82 | 1.51 | 25 | 4 | 37.5 |
| tracker, ceiling 1.25 | 121 | 104 | 41 | 1.26 | 1.30 | 25 | 4 | 25.2 |
| tracker, ceiling 1.5 | 116 | 104 | 42 | 0.94 | 1.21 | 25 | 4 | 11.0 |
| daily, ceiling 1.25 | 106 | 104 | 44 | 1.45 | 1.22 | 25 | 4 | 33.0 |
| SPOT, ceiling 1.25 | 106 | 104 | 43 | 1.41 | 0.84 | 25 | 4 | 35.6 |
| SPOT, approval every 6 h | 106 | 104 | 43 | 1.42 | 0.84 | 25 | 4 | 31.7 |
| SPOT, approval daily | 106 | 104 | 43 | 1.42 | 0.84 | 25 | 4 | 31.7 |

# F Exploratory round after the second review (tt1f)

After the registered finals of tt1d, two mock reviews of this paper and an audit of the analysis code led to one more round of analyses on data that had already had its registered access. A written plan was committed before the runs, with two dated amendments, each committed before the analyses it added; every new access to attack data or sealed runs is in the ledger with a reason that begins with `EXPLORATORY`. Nothing in this appendix is registered.

*Chance reference.* Table 11 in Section 6.2 gives, for each rule, the events it detected on Tennessee Eastman and the number its alarms detect when the alarm sequence of each run is shifted circularly by a random offset (200 shifts per run). Under the onset count the noise followers are detected below chance by both the frozen threshold (6 against 23) and the tracker (1 against 26): the alarm is already running, or the rule is raised. On BATADAL the frozen threshold's 14 of 14 compare with 5.7 [2, 10] and the $\tau_0$-scaled tracker's 14 with 4.3 [1, 8] (1000 shifts per file).

*Time to clear.* In the 36 noise sequences (median length 41.4 h) each of the five rules below alarmed in every sequence, at a median of one step after its start. The share of the sequence spent in alarm (median) was 100% for the frozen threshold, 37% for the tracker, 56% for the slow tracker, 32% for the tracker with $\eta = 0.01\,\tau_0$ and 43% for daily recalibration; their last alarms came a median 23.7, 27.6, 22.6 and 22.9 h into the sequence.

*Peak scores.* Of the 245 Tennessee Eastman events, 25 never exceed $\tau_0$, 61 never exceed $1.1\,\tau_0$ and 70 never exceed $1.25\,\tau_0$; the median peak is $17\,\tau_0$. All 70 of the latter are first events. The 14 BATADAL attacks peak between 3.2 and 1,583 $\tau_0$.

*Code audit.* An audit of the analysis code after the finals found five defects in files frozen by the registrations. The files stay as registered; corrected versions and regression tests are in the artifact beside them. (1) The calibration search returns the smallest candidate threshold that meets the false-alarm budget, although the number of alarm episodes is not monotone in the threshold; the corrected search stops at the first candidate that breaks the budget. On the seven Tennessee Eastman monitors and the three HAI development releases it returns the D1 threshold of the frozen search exactly; HAI 20.07, BATADAL, ProvICS and the side scores were not rechecked. (2) The ProvICS loader marks the steps inside a recording gap invalid and leaves valid the 36 steps after it, whose scores still depend on filled values (Section 6.4). (3) The first smoothing window of a stream mixes in later samples (at most 49 scored steps on HAI, 6 on Tennessee Eastman, 19 on ProvICS). (4) The online daily rule reaches into the previous day after invalid steps; no registered stream has such steps outside its warm-up. (5) The online episode counter of the hold-off tracker splits alarm runs separated by exactly the merging gap. The audit also notes that the event bootstrap of tt1 and tt1b treats events of one stream as independent.

**Table 22: SPOT under the 24 h frog on the 56 sealed Tennessee Eastman placements: bumps of 1.05 / 1.25 / 1.5 / 2 $\tau_0$ caught after the campaign and in the control, the threshold over $\tau_0$ and the fitted shape $\xi$ at the end of the campaign (quartiles), and placements in which the campaign raised no alarm of the attacked rule. Attack-free shapes lie between $-0.16$ and $0.35$. Data status: the caught and silent counts of the first row are registered (tt1d) and reproduced; everything else is exploratory (tt1f).**

| Rule | after the frog | control | $\tau/\tau_0$ at the end | $\xi$ at the end | silent |
|---|---|---|---|---|---|
| standard SPOT (registered) | 0 / 16 / 24 / 32 | 44 / 56 / 56 / 56 | 1.24, 1.75, $10^9$ | -1.22, -1.07, 6.53 | 51 |
| shape bounded to 0.5 | 0 / 0 / 0 / 21 | 44 / 56 / 56 / 56 | 1.80, 2.26, 34.78 | -0.50, -0.50, 0.50 | 52 |
| runs declustering | 44 / 56 / 56 / 56 | 43 / 56 / 56 / 56 | 0.99, 1.02, 1.05 | -0.11, 0.03, 0.20 | 25 |

**Table 23: Sensor-level campaigns on normal data of the three spent HAI releases (30 placements). The attacker writes three sensor readings and the real monitor scores the result; the plant's response is not simulated. Terminal setpoint steps of 1 / 1.5 / 2 / 3 times the smallest step the frozen threshold detects, caught after the campaign and in the control; placements in which the campaign raised no alarm of the attacked rule; alarm episodes during the campaign and the threshold over $\tau_0$ at the terminal step (medians). Data status: exploratory (tt1f).**

| Campaign | Rule | after | control | silent | episodes | $\tau/\tau_0$ |
|---|---|---|---|---|---|---|
| frog, reads $\tau_t$ (6 h) | frozen | 30 / 30 / 30 / 30 | 30 / 30 / 30 / 30 | 15 | 0.5 | 1.00 |
| frog, reads $\tau_t$ (6 h) | standard SPOT | 9 / 21 / 30 / 30 | 30 / 30 / 30 / 30 | 12 | 1 | 1.11 |
| frog, sees only alarms (6 h) | frozen | 30 / 30 / 30 / 30 | 30 / 30 / 30 / 30 | 0 | 13.5 | 1.00 |
| frog, sees only alarms (6 h) | standard SPOT | 20 / 30 / 30 / 30 | 30 / 30 / 30 / 30 | 0 | 13 | 1.03 |
| 20 s pulses every 30 min (24 h) | frozen | 30 / 30 / 30 / 30 | 30 / 30 / 30 / 30 | 0 | 49 | 1.00 |
| 20 s pulses every 30 min (24 h) | daily, episodes left out | 21 / 30 / 30 / 30 | 30 / 30 / 30 / 30 | 0 | 50 | 1.03 |
| 20 s pulses every 30 min (24 h) | tracker, hold-off | 0 / 0 / 0 / 1 | 21 / 28 / 30 / 30 | 0 | 48 | 12.45 |